

Future Work: Dual-Trace Demand–Supply Modulation for QD³

Proposed extension preserving the current Oja-style low-rank query dynamics.

Motivation. The current QD³ uses historical reads to estimate where memory accuracy matters and increases the residual-correction rate when a later write arrives in a highly queried direction. A natural extension is to maintain a second, structurally matched trace of write history. The goal is not simply to add read and write heat, but to estimate a *demand–supply mismatch*: addresses with high read demand but relatively few previous write opportunities should receive the strongest extra correction when a new write finally arrives.

1. Read-demand trace: retain the current QD³ mechanism. For query $q_j \in \mathbb{R}^{d_k}$, let $U_j^{(r)} = [u_{j,1}^{(r)}, \dots, u_{j,r}^{(r)}] \in \mathbb{R}^{d_k \times r}$ and $a_j^{(r)} \in \mathbb{R}_+^r$. The update is exactly the same Oja-style low-rank construction as in the current method:

$$z_j^{(r)} = (U_{j-1}^{(r)})^T q_j, \\ U_j^{(r)} = U_{j-1}^{(r)} + \eta_r (q_j - U_{j-1}^{(r)} z_j^{(r)}) (z_j^{(r)})^T, \quad a_j^{(r)} = \lambda_r a_{j-1}^{(r)} + z_j^{(r)} \odot z_j^{(r)}.$$

For an incoming write key k_t , the read heat is

$$\bar{h}_r(k_t) = \sum_{i=1}^r a_{j,i}^{(r)} ((u_{j,i}^{(r)})^T k_t)^2, \quad \hat{h}_r(k_t) = \bar{h}_r(k_t) / s_r,$$

where $j = j(t)$ is the latest query state and s_r is a causal normalization scale, analogous to the current QD³ normalization.

2. Write-supply trace: mirror the same low-rank dynamics. Maintain a second state $U_t^{(w)}, a_t^{(w)}$ with the same form. Crucially, the current write should *not* count as previous supply before its own priority is computed. Therefore the incoming key is first evaluated using the pre-write trace:

$$\bar{h}_w(k_t) = \sum_{i=1}^r a_{t-1,i}^{(w)} ((u_{t-1,i}^{(w)})^T k_t)^2, \quad \hat{h}_w(k_t) = \bar{h}_w(k_t) / s_w.$$

After the memory write, the write trace is updated with the same Oja mechanism:

$$z_t^{(w)} = (U_{t-1}^{(w)})^T k_t, \\ U_t^{(w)} = U_{t-1}^{(w)} + \eta_w (k_t - U_{t-1}^{(w)} z_t^{(w)}) (z_t^{(w)})^T, \quad a_t^{(w)} = \lambda_w a_{t-1}^{(w)} + z_t^{(w)} \odot z_t^{(w)}.$$

Thus both read demand and write supply are represented by the same compact, adaptive low-rank geometry rather than by dense $d_k \times d_k$ moments.

3. Demand–supply priority and Delta update. After causal normalization of the two heats, define

$$p_{\text{raw}}(k_t) = \frac{\hat{h}_r(k_t)}{\epsilon + \hat{h}_w(k_t)}, \quad p(k_t) = \frac{p_{\text{raw}}(k_t)}{1 + p_{\text{raw}}(k_t)}.$$

The bounded priority can modulate the same residual Delta correction used by QD³:

$$\tilde{\eta}_t = \min\{\eta(1 + \beta p(k_t)), c\}, \quad M_t = M_{t-1} + \tilde{\eta}_t (v_t - M_{t-1} k_t) k_t^T.$$

With $\beta = 0$, the operator again reduces exactly to ordinary Delta (assuming $\eta \leq c$). Choosing $c < 2$ preserves the same written-address non-expansion interval for unit-norm keys.

Read demand	Write supply	Priority	Interpretation
High	Low	Very high	Important but under-served
High	High	Moderate	Important and already well updated
Low	High	Low	Frequently written but rarely needed
Low	Low	Low	Little evidence of importance

4. Expected value and evaluation. This extension reframes QD³ as an online *demand–supply allocator*: the read trace estimates where accuracy is valuable, while the write trace estimates where correction opportunities have already been abundant. The main hypothesis is that the dual-trace version should help most when $p_{\text{query}} \neq p_{\text{write}}$, especially for addresses that are queried often but written rarely. A future study should compare read-only QD³, write-only trace, simple read+write fusion, the demand/supply ratio above, and full-moment versus low-rank dual traces. Uniform or closely matched read/write distributions should provide an important negative control, where little additional gain is expected.